Science Mock Exam

Foundational biology, chemistry, physics, and Earth science.

70 original multiple-choice practice questions ? Choose the best answer for each item.

Unofficial practice material created for ScholarHunt. This is not an official scholarship examination.

1. Which organ pumps blood through the human body?

A. Lungs
B. Heart
C. Stomach
D. Kidneys

2. What is the main source of energy for Earth’s climate system?

A. The Moon
B. The Sun
C. Volcanoes
D. Ocean tides

3. Which state of matter has a fixed volume but no fixed shape?

A. Solid
B. Liquid
C. Gas
D. Plasma only

4. Which gas do humans need for cellular respiration?

A. Oxygen
B. Helium
C. Neon
D. Methane

5. What is the basic unit of life?

A. Atom
B. Cell
C. Organ
D. Tissue

6. Which material is attracted to a magnet?

A. Iron
B. Glass
C. Paper
D. Wood

7. What force pulls objects toward Earth?

A. Friction
B. Gravity

C. Magnetism
D. Buoyancy

8. Which planet is closest to the Sun?

A. Venus
B. Earth
C. Mercury
D. Mars

9. What process changes liquid water into water vapor?

A. Freezing
B. Condensation
C. Evaporation
D. Melting

10. Which part of a plant makes most of its food?

A. Root
B. Leaf
C. Flower
D. Seed coat

11. Which body system helps protect the body from many infections?

A. Immune system
B. Skeletal system
C. Digestive system
D. Muscular system

12. What is the chemical symbol for water?

A. CO2
B. H2O
C. O2
D. NaCl

13. Which is a renewable energy source?

A. Coal
B. Natural gas
C. Sunlight
D. Petroleum

14. Which layer of Earth is the outermost solid layer?

A. Inner core
B. Outer core
C. Mantle
D. Crust

15. What do we call an animal that eats both plants and animals?

A. Herbivore

B. Carnivore
C. Omnivore
D. Producer

16. Which instrument measures temperature?

A. Barometer
B. Thermometer
C. Anemometer
D. Compass

17. What happens to most materials when heated?

A. They contract
B. They expand
C. They disappear
D. They lose all mass

18. Which is an example of a chemical change?

A. Ice melting
B. Paper being cut
C. Iron rusting
D. Water freezing

19. What is the role of roots in most plants?

A. Absorb water and anchor the plant
B. Make seeds only
C. Attract pollinators
D. Release sound

20. Which object reflects most visible light?

A. A clean mirror
B. Black cloth
C. Dark soil
D. Charcoal

21. What is the closest star to Earth?

A. Sirius
B. The Sun
C. Polaris
D. Alpha Centauri A

22. Which blood cells help carry oxygen?

A. Red blood cells
B. Platelets
C. White blood cells
D. Nerve cells

23. What is the main function of the lungs?

A. Pump blood
B. Exchange oxygen and carbon dioxide
C. Digest food
D. Filter urine

24. Which habitat is characterized by very little rainfall?

A. Desert
B. Rainforest
C. Wetland
D. Coral reef

25. What is the name for a scientist who studies weather?

A. Geologist
B. Meteorologist
C. Astronomer
D. Botanist

26. Which part of an atom has a negative electrical charge?

A. Proton
B. Neutron
C. Electron
D. Nucleus

27. What happens during condensation?

A. Gas changes to liquid
B. Liquid changes to solid
C. Solid changes to gas
D. Liquid changes to gas

28. Which is the best example of a simple machine?

A. Lever
B. Battery
C. Light bulb
D. Thermometer

29. Which organisms break down dead plants and animals?

A. Decomposers
B. Producers
C. Predators
D. Pollinators

30. Which planet has prominent rings that are easily seen from Earth with a telescope?

A. Mars
B. Saturn
C. Mercury
D. Venus

31. What is the primary function of the skeleton?

A. Support and protect the body
B. Digest nutrients
C. Make hormones only
D. Exchange gases

32. Which type of rock forms when magma or lava cools?

A. Sedimentary
B. Igneous
C. Metamorphic
D. Fossil

33. What is the term for the path a planet follows around a star?

A. Axis
B. Orbit
C. Equator
D. Crater

34. Which nutrient is the body’s main quick source of energy?

A. Carbohydrate
B. Vitamin C
C. Water
D. Calcium

35. Which statement about sound is correct?

A. Sound travels through a vacuum
B. Sound needs a medium to travel
C. Sound is always visible
D. Sound travels only through metal

36. What is the largest organ of the human body?

A. Heart
B. Skin
C. Liver
D. Lung

37. Which natural event is caused by movement along faults in Earth’s crust?

A. Earthquake
B. Eclipse
C. Tide
D. Drought

38. What do bees commonly transfer between flowers during pollination?

A. Pollen
B. Roots
C. Seeds only
D. Water vapor

39. Which part of the water cycle forms clouds from cooled water vapor?

A. Condensation
B. Infiltration
C. Runoff
D. Collection

40. Which planet is known for its large red storm called the Great Red Spot?

A. Jupiter
B. Earth
C. Neptune
D. Mercury

41. Which is a producer in a food chain?

A. Grass
B. Hawk
C. Frog
D. Mushroom

42. What is the usual effect of friction on moving objects?

A. It resists motion
B. It creates mass
C. It removes gravity
D. It changes matter into light

43. Which vitamin can the body make in skin with adequate sunlight exposure?

A. Vitamin A
B. Vitamin C
C. Vitamin D
D. Vitamin K only

44. What is the boiling point of pure water at sea level in Celsius?

A. 0 degrees
B. 50 degrees
C. 100 degrees
D. 212 degrees

45. Which instrument is used to view very small cells?

A. Microscope
B. Telescope
C. Periscope
D. Binoculars

46. What is biodiversity?

A. The variety of living things
B. The movement of clouds
C. The age of a rock
D. The depth of an ocean

47. Which gas makes up the largest share of Earth’s atmosphere?

A. Oxygen
B. Nitrogen
C. Carbon dioxide
D. Argon

48. Which material is a good electrical conductor?

A. Copper
B. Rubber
C. Dry wood
D. Glass

49. What do we call the change from solid ice to liquid water?

A. Freezing
B. Melting
C. Condensation
D. Sublimation

50. Which part of a flower produces pollen in many flowering plants?

A. Anther
B. Petal
C. Sepal
D. Ovary

51. Which body system breaks food into absorbable nutrients?

A. Digestive system
B. Respiratory system
C. Nervous system
D. Skeletal system

52. What is the main reason Earth has seasons?

A. Its tilted axis as it orbits the Sun
B. Its distance from the Moon
C. Changes in the Sun’s size
D. Clouds blocking sunlight

53. Which force keeps planets in orbit around the Sun?

A. Gravity
B. Friction
C. Sound
D. Electric current

54. What is a fossil?

A. Preserved evidence of ancient life
B. A newly formed mineral
C. A type of cloud
D. A modern seed

55. Which is a physical change?

A. Sugar dissolving in water
B. Wood burning
C. Milk souring
D. Iron rusting

56. Which part of the eye controls how much light enters?

A. Iris
B. Retina
C. Optic nerve
D. Cornea only

57. What is the main purpose of a vaccine?

A. Train immune defenses to recognize a pathogen
B. Cure every illness immediately
C. Replace all white blood cells
D. Stop the heart from infection

58. Which celestial body naturally orbits Earth?

A. The Moon
B. Mars
C. The Sun
D. Venus

59. Which process releases water vapor from plant leaves?

A. Transpiration
B. Pollination
C. Germination
D. Digestion

60. What is the unit of electric current?

A. Ampere
B. Meter
C. Liter
D. Celsius

61. Which animal is an invertebrate?

A. Butterfly
B. Lizard
C. Frog
D. Eagle

62. What kind of energy is stored in a stretched elastic band?

A. Potential energy
B. Sound energy
C. Nuclear energy
D. Light energy

63. Which Earth layer is mostly liquid metal?

A. Crust
B. Mantle
C. Outer core
D. Inner core

64. Which is an example of adaptation?

A. A cactus has spines that reduce water loss
B. A rock changes color in shade
C. A river flows downhill
D. Ice melts in sunlight

65. What is the role of a control group in a fair experiment?

A. Provide a comparison without the tested change
B. Guarantee the hypothesis is correct
C. Change every variable
D. Replace all measurements

66. Which molecule carries hereditary information in most organisms?

A. DNA
B. Starch
C. Water
D. Oxygen

67. Which type of energy does a moving bicycle have?

A. Kinetic energy
B. Chemical energy only
C. Nuclear energy
D. Elastic energy only

68. What does a thermometer reading in Celsius describe?

A. Temperature
B. Mass
C. Distance
D. Time

69. Which natural resource is replenished by the water cycle?

A. Fresh water
B. Coal
C. Petroleum
D. Metal ore

70. Which part of the digestive system absorbs most nutrients?

A. Small intestine
B. Esophagus
C. Mouth
D. Large intestine

Answer Key & Explanations

Use this section after completing the practice exam.

1. B ? Heart
The heart contracts to move blood through blood vessels.

2. B ? The Sun
Sunlight supplies most energy driving weather and climate.

3. B ? Liquid
A liquid keeps its volume but takes its container’s shape.

4. A ? Oxygen
Cells use oxygen to release energy from food.

5. B ? Cell
A cell is the smallest unit considered living.

6. A ? Iron
Iron is a ferromagnetic material attracted to magnets.

7. B ? Gravity
Earth’s gravity attracts objects toward its center.

8. C ? Mercury
Mercury is the innermost planet in the Solar System.

9. C ? Evaporation
Evaporation changes liquid water into a gas.

10. B ? Leaf
Leaves capture light for photosynthesis.

11. A ? Immune system
The immune system recognizes and helps fight pathogens.

12. B ? H2O
A water molecule has two hydrogen atoms and one oxygen atom.

13. C ? Sunlight
Sunlight is continually replenished on human timescales.

14. D ? Crust
The crust is Earth’s thin outer rocky layer.

15. C ? Omnivore
Omnivores obtain food from plant and animal sources.

16. B ? Thermometer
A thermometer measures temperature.

17. B ? They expand
Heating often makes particles move more and spread apart.

18. C ? Iron rusting
Rusting forms new substances from iron and oxygen.

19. A ? Absorb water and anchor the plant
Roots commonly anchor plants and absorb water and minerals.

20. A ? A clean mirror
A smooth mirror reflects light in an orderly way.

21. B ? The Sun
The Sun is the star at the center of our Solar System.

22. A ? Red blood cells
Hemoglobin in red blood cells carries much of the blood’s oxygen.

23. B ? Exchange oxygen and carbon dioxide
The lungs exchange gases between air and blood.

24. A ? Desert
Deserts receive little precipitation compared with many habitats.

25. B ? Meteorologist
Meteorologists study the atmosphere and weather.

26. C ? Electron
Electrons carry negative charge.

27. A ? Gas changes to liquid
Condensation occurs when vapor becomes liquid.

28. A ? Lever
A lever is one of the classical simple machines.

29. A ? Decomposers
Decomposers recycle nutrients from dead organic matter.

30. B ? Saturn
Saturn’s extensive rings are especially prominent.

31. A ? Support and protect the body
The skeleton supports the body and protects organs.

32. B ? Igneous
Cooling molten rock produces igneous rock.

33. B ? Orbit
An orbit is the path followed by an object around another.

34. A ? Carbohydrate
The body commonly breaks down carbohydrates for energy.

35. B ? Sound needs a medium to travel
Sound is a vibration that travels through matter.

36. B ? Skin
The skin covers and protects the body and is its largest organ.

37. A ? Earthquake
Fault movement can release energy as seismic waves.

38. A ? Pollen
Bees carry pollen between flowers as they forage.

39. A ? Condensation
Water vapor cools and condenses into tiny droplets or ice crystals.

40. A ? Jupiter
The Great Red Spot is a long-lasting storm on Jupiter.

41. A ? Grass
Grass makes food using sunlight through photosynthesis.

42. A ? It resists motion
Friction acts against relative motion between surfaces.

43. C ? Vitamin D
Sunlight helps skin produce vitamin D.

44. C ? 100 degrees
At standard sea-level pressure, water boils near 100 degrees Celsius.

45. A ? Microscope
A microscope magnifies objects too small to see clearly unaided.

46. A ? The variety of living things
Biodiversity describes the variety of life in an area or on Earth.

47. B ? Nitrogen
Nitrogen makes up about 78 percent of dry air.

48. A ? Copper
Copper allows electric charge to move readily.

49. B ? Melting
Melting changes a solid into a liquid.

50. A ? Anther
The anther produces pollen grains.

51. A ? Digestive system
The digestive system processes food and absorbs nutrients.

52. A ? Its tilted axis as it orbits the Sun
Earth’s axial tilt changes sunlight angle and day length through the year.

53. A ? Gravity
The Sun’s gravity helps keep planets in their orbits.

54. A ? Preserved evidence of ancient life
Fossils preserve remains or traces of past organisms.

55. A ? Sugar dissolving in water
Dissolving does not create a new substance; the sugar remains sugar.

56. A ? Iris
The iris changes pupil size to regulate incoming light.

57. A ? Train immune defenses to recognize a pathogen
Vaccines prepare immune responses to a specific disease threat.

58. A ? The Moon
The Moon is Earth’s natural satellite.

59. A ? Transpiration
Plants release water vapor through small openings in their leaves.

60. A ? Ampere
Electric current is measured in amperes.

61. A ? Butterfly
A butterfly has no backbone.

62. A ? Potential energy
Its stretched form stores elastic potential energy.

63. C ? Outer core
The outer core is primarily liquid iron and nickel.

64. A ? A cactus has spines that reduce water loss
Spines help many cacti limit water loss and provide defense.

65. A ? Provide a comparison without the tested change
A control offers a baseline for comparison.

66. A ? DNA
DNA stores genetic instructions in most living organisms.

67. A ? Kinetic energy
Kinetic energy is energy of motion.

68. A ? Temperature
Celsius is a scale for measuring temperature.

69. A ? Fresh water
Evaporation and precipitation continually cycle water.

70. A ? Small intestine
Most nutrient absorption takes place in the small intestine.